

HDOC DAY-36

Title: Modular String Processing without Built-in Library Functions

1. Problem Statement

Write a menu-driven C program to perform the following operations using separate functions:

1. Count characters in a string without using built-in length functions.
2. Print each character of a string on a new line.
3. Count vowels and consonants in a string.
4. Exit.

2. Theoretical Background & Logic

1. Null-Terminator Handling (`\0`):

In C, strings are null-terminated character arrays. Custom string length iteration traverses indices $i = 0, 1, 2, \dots$ until reaching `str[i] == '\0'`.

2. Character Classification:

- Alphabet check: `'a' <= character <= 'z'` (after normalizing to lowercase using `tolower()`).
- Vowels: Matches `'a', 'e', 'i', 'o', or 'u'`.

- Consonants: Any alphabet character that is not a vowel.

3. Test Case Table

Test Case	Selected Option	Input String	Expected Output	Actual Output	Status
1	Count Characters	"Hello"	Length = 5	Length = 5	Pass
2	Print Newline	"Cat"	C		

a

t | C

a

t | Pass | | **3** | Vowels/Consonants | "HDOC Day 36" |
Vowels: 2 (O, a)

Consonants: 5 (H, D, C, D, y) | Vowels: 2

Consonants: 5 | Pass | | **4** | Exit | Option 5 | Terminate clean
| Terminated clean | Pass |

4. Source Code Listing

```
#include <ctype.h>
#include <stdio.h>

void read_string(char str[], int max_size) {
    printf("Enter a string: ");
    scanf(" %[^\\n]", str);
}

int count_characters(const char str[]) {
    int length = 0;
    while (str[length] != '\\0') {
        length++;
    }
    return length;
}

void print_characters_newline(const char str[]) {
    printf("\\nCharacters on individual lines:\\n");
    int i = 0;
```

```

while (str[i] != '\0') {
    printf("%c\n", str[i]);
    i++;
}
}

void count_vowels_and_consonants(const char str[]) {
    int vowels = 0, consonants = 0, digits = 0, special = 0;
    int i = 0;

    while (str[i] != '\0') {
        char ch = tolower((unsigned char)str[i]);
        if (ch >= 'a' && ch <= 'z') {
            if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u') {
                vowels++;
            } else {
                consonants++;
            }
        } else if (ch >= '0' && ch <= '9') {
            digits++;
        } else {
            special++;
        }
        i++;
    }

    printf("\n--- Vowel and Consonant Analysis ---\n");
    printf("Vowels    : %d\n", vowels);
    printf("Consonants : %d\n", consonants);
    printf("Digits    : %d\n", digits);
    printf("Special/Spaces : %d\n", special);
}

int main(void) {
    char str[200];
    int choice;

    read_string(str, 200);

    do {

```

```

printf("\n===== STRING OPERATIONS MENU
=====\\n");
printf("1. Count characters in string (without built-in functions)\\n");
printf("2. Print each character on a new line\\n");
printf("3. Count vowels and consonants\\n");
printf("4. Enter a new string\\n");
printf("5. Exit\\n");
printf("Enter your choice (1-5): ");
scanf("%d", &choice);

switch (choice) {
    case 1:
        printf("\\nTotal string length (character count): %d\\n",
count_characters(str));
        break;
    case 2:
        print_characters_newline(str);
        break;
    case 3:
        count_vowels_and_consonants(str);
        break;
    case 4:
        read_string(str, 200);
        break;
    case 5:
        printf("Exiting program...\\n");
        break;
    default:
        printf("Error: Invalid choice! Please select between 1 and 5.\\n");
        break;
}
} while (choice != 5);

return 0;
}

```

5. Execution Screenshots

```
kali@kali:~/HDOOC_Day36
Session Actions Edit View Help
(kali@kali)~$ mkdir HDOOC_Day36
cd HDOOC_Day36
(kali@kali)~/HDOOC_Day36$ nano day36.c
(kali@kali)~/HDOOC_Day36$ gcc day36.c -o day36
./day36
Enter a string: Hello World 123

===== STRING OPERATIONS MENU =====
1. Count characters in string (without built-in functions)
2. Print each character on a new line
3. Count vowels and consonants
4. Enter a new string
5. Exit
Enter your choice (1-5): 1

Total string length (character count): 15

===== STRING OPERATIONS MENU =====
1. Count characters in string (without built-in functions)
2. Print each character on a new line
3. Count vowels and consonants
4. Enter a new string
5. Exit
Enter your choice (1-5): 2

Characters on individual lines:
H
e
l
l
o

W
o
r
l
d

1
2
3

===== STRING OPERATIONS MENU =====
1. Count characters in string (without built-in functions)
2. Print each character on a new line
3. Count vowels and consonants
4. Enter a new string
5. Exit
Enter your choice (1-5): █
```